Tutorial 4: Data Flow Diagram (DFD)

Today, you will learn DFD using a simple Library Management System example.

But your Lab 4 submission will be based on YOUR OWN PROJECT.

Use the requirements and functions already identified for your project.

Main idea: Learn with one example → apply the same steps to your own project.

How Lab 4 connects with your project

Your project already has users and functions.

A DFD shows how DATA moves between users, processes and stored information.

Continue with the same project.

Connection with Earlier Labs

Lab 1: Requirements/SRS helped you understand what a system should do.

Lab 2: Process model helped you understand how software may be developed.

Lab 3: Use Case Diagram showed who uses the system and what they do.

Lab 4: DFD will show how data moves inside YOUR project.

What is a DFD?

DFD means Data Flow Diagram.

It shows where data comes from, where it goes, how it is processed, and where it is stored.

A DFD focuses on DATA, not program code.

Example: Student sends Login Details → System checks details → Login Result is returned.

Four basic parts of a DFD

1. External Entity – person/system outside your project that sends or receives data.
2. Process – work done by the system.
3. Data Flow – data moving from one place to another.
4. Data Store – place where data is saved.

1. External Entity

An external entity is outside the software system.

It gives data to the system or receives data from the system.

Examples depend on the project: Student, Customer, Admin, Doctor, Librarian, Teacher.

Choose entities from YOUR project, not from a copied diagram.

2. Process

A process changes input data into useful output.

Use action words for process names.

Examples: Verify Login, Manage Orders, Issue Book, Generate Report, Record Attendance.

Your processes should come from the functions/requirements of your project.

3. Data Flow

A data flow is the information moving between DFD elements.

Show it using an arrow.

Write the name of the DATA on the arrow.

Good labels: Login Details, Search Request, Payment Details, Report.

Avoid labels such as 'click', 'go', 'next' because these are actions, not data.

4. Data Store

A data store keeps information for later use.

Examples: User Database, Book Records, Order Records, Student Records.

Only add data stores actually needed by YOUR project.

An external entity should not directly connect to a data store.

Context Diagram

The whole software project is shown as ONE process.

Show only the main external entities and major input/output data.

Do not show internal processes or data stores here.

Write your own project name inside the main process.

Example project name: Online Food Ordering System / College Portal / Library System.

Context Diagram – simple example

Example only: Library Management System

Member → Search Request → Library Management System

Library Management System → Search Result → Member

Librarian → Book/Member Details → Library Management System

Library Management System → Transaction/Report Information → Librarian

Students must replace this example with entities and data from their own project.

Level 0 DFD

Now break the whole system into 3–6 major processes.

Add important data stores.

Connect entities, processes and data stores using labelled data arrows.

The processes must represent major functions of YOUR project.

Level 0 – example

Example only: Library Management System

1.0 Manage Members

2.0 Manage Books

3.0 Manage Loans

Possible data stores: Member Records, Book Records, Loan Records.

For another project, the process names will be different.

How to make Level 0 for YOUR project

Step 1: Open your project requirements/use cases.

Step 2: Select the major functions.

Step 3: Convert major functions into DFD processes.

Step 4: Identify the data each process needs.

Step 5: Identify where that data is stored.

Step 6: Connect everything using labelled arrows.

Level 1 DFD

Choose ONE important process from your Level 0 DFD.

Break that process into smaller steps.

Example: 'Manage Orders' may become Receive Order, Check Items, Confirm Order, Update Order Record.

Your Level 1 processes must still belong to the selected Level 0 process.

Balancing – very simple meaning

Data entering/leaving a process at one level should remain consistent when that process is expanded.

Do not suddenly add an unrelated external input/output in Level 1.

Check that your Context Diagram, Level 0 and Level 1 tell the same story.

Think: same system, only more detail at each next level.

Project examples

Food Delivery Project → Customer, Restaurant, Admin; Orders, Payment, Menu data.

Attendance System → Student, Faculty, Admin; Attendance and Report data.

Hospital System → Patient, Doctor, Receptionist; Patient and Appointment data.

College Portal → Student, Faculty, Admin; Course, Result and Registration data.

Your own project may use completely different entities and processes.

Common mistakes

Copying a Library Management DFD even when your project is different.

Using functions that are not present in your project requirements.

Showing control flow instead of data flow.

Connecting an external entity directly to a data store.

Using different input/output data at different DFD levels without reason.

Making the diagram too complicated.

What you will do in Lab 4

Use YOUR selected project.

Write a short project description.

List external entities and major data.

Draw one Context Diagram.

Draw one Level 0 DFD.

Choose one Level 0 process and draw its Level 1 DFD.

Check balancing and write a short result.

Library Management System may be used as a classroom example, not as everyone's compulsory submission.

Quick class activity

Take your own project title.

Write any 2 external entities.

Write any 3 major processes.

Write any 2 data stores.

Give one example of input data and one example of output data.

Share your answer with the class.